

Main Idea:

The main goal of this project was to build a solid and reliable backend system for an online store. We wanted to create a single place that handles both the store management side and the customer shopping experience. On the management side, we built tools to help admins run the company, such as a way to add or remove staff members, tools to keep track of products in the inventory, and a system to manage customer orders as they come in. On the customer side, we focused on making the shopping process easy and secure. We added features that allow users to sign up for accounts, handle payments safely through Stripe, save their order history so they can look back at past purchases, and automatically send emails to keep them updated on their transactions. Overall, the system works as a complete tool to make sure that the staff can manage the shop easily while the customers have a smooth and safe experience.

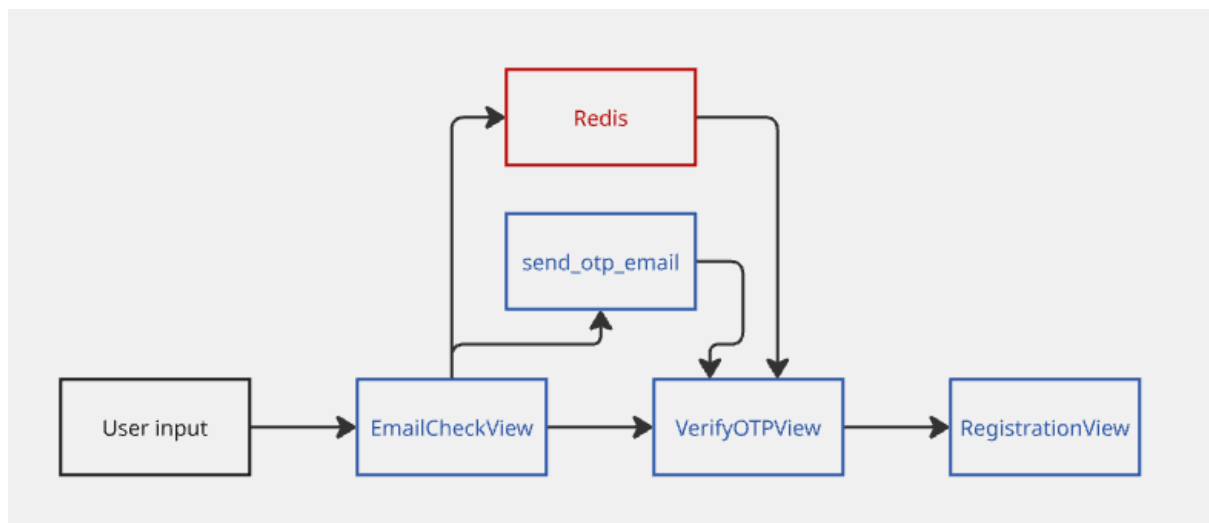
Technology Stack:

- Python 3.14
- Django
- PostgreSQL 17v
- Redis 8.6.3 + RedisInsight
- NextJS 16
- Tailwind 4v
- ShadCN
- Docker

Ports:

- Api: 8000
- Redis: 6379/0
- Admin frontend: 3000
- Client frontend: 3001

Registration Logic Example:



When user writes email, EmailCheckView validates email and ,if its new user, send OTP code on email:

Confirmation email

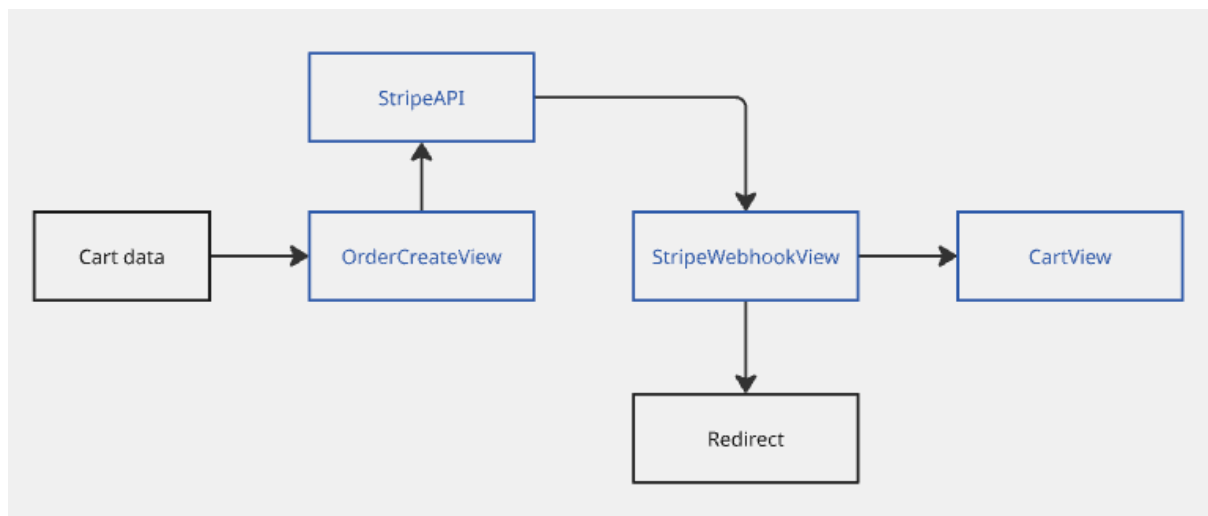
Use the code below to complete the verification.

865495

The code expires in 5 minutes. If you didn't request this code, simply ignore the email.

Then OTP code is validated by VerifyOTPView and allows the user to write a password. Then the user writes 2 passwords and if they are equal the user gets JWT.

Order Logic Example:



If user data in Cart, he can make an order. He writes order form

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< BACK TO CART

CHECKOUT

SHIPPING ADDRESS

FIRST NAME * LAST NAME *

CITY * POSTAL CODE *

ADDRESS *

ORDER SUMMARY

ITEMS (3) 100,000 KZT

TOTAL 100,000 KZT

[PLACE ORDER](#)

and this data goes to OrderCreateView and redirects request to `payment-success?order_id={order.id}`. If payment is successful, order writes into DB and we can see our income in Stripe:

